



Modified imperialist competitive algorithm for aircraft landing scheduling problem

Kimia Shirini¹ · Hadi S. Aghdasi¹ · Saeed Saeedvand²

Accepted: 14 February 2024 / Published online: 12 March 2024

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Abstract

In recent years, airport runways have become a more critical bottleneck in airports, and it is very unusual to use only one runway to solve the Aircraft Landing Problem (ALP). The ALP includes the aircraft's landing scheduling and assigning them to runways. In addition to certain limited time frames for aircraft while landing, to prevent accidents, the distance between aircraft should be restricted during the flight and in the landing phase. In this paper, to solve the problem in multi-runway mode, a solution is proposed that has considered all the limitations to create a trade-off between the runways to reduce the traffic in the runways. The present study considers the balance between bands and offers a new method of improving the Imperialist Competitive Algorithm (ICA) while reducing the cost due to the early and late landing of the aircraft. In other words, a novel approach for addressing the ALP with multiple criteria, employing a delay and early landing cost optimization technique and runway balance strategy, as well as using multi-runway, which will reflect the current realities of the aviation industry and provide a more accurate and relevant analysis, has been presented. Thirty-two benchmark instances were selected and compared with four famous algorithms: Particle Swarm Optimization (PSO), Immunoglobulin-Based Artificial Immune System (IAIS), Grey Wolf Optimizer (GWO), and flower pollination algorithm (FPA) as the results on a small-scale indicate, the ICA method performs superior outcomes, except for one case. Regarding the two objectives, the presented method, compared to other methods, managed to reduce the cost by 3.5% (on a small-scale). Furthermore, on a large scale (500 aircraft), improved ICA has been able to reduce the cost of the early or late arrival of the aircraft by 35%.

Keywords Aircraft landing scheduling · Imperialist competitive algorithm · Runway balance · Metaheuristic method · Multi-objective problem

1 Introduction

Congestions in airport runways are considered the bottlenecks of air traffic control [1, 2]. Despite their role in connecting air routes and ground traffic within the airports, the runway system is becoming a significant problem for airport operators [3]. The demand for air travel has always been on the rise, and the existence of low-cost airlines in Western nations during the recent decades has boosted it up. As the Federal Aviation Administration reports, airlines in the next couple of decades will witness 100–150% increase [4]. The total annual number of flights has risen from 6.1 million in 2006 to 71.2 million in 2022 and is expected to reach approximately 95.9 million in 2030 [5]. During the COVID-19 pandemic, the aviation industry faced an unprecedented reduction in the number of flights worldwide [6]. At its peak in 2020, global air traffic experienced a reduction of around 60–70% compared to pre-pandemic levels, which forced the total number of flights worldwide down to 22.2 million in 2021. However, it increased again in 2022 and reached the rate of 30 million flights. According to estimations, there is going to be a 50% increase in the number of passengers by 2025 [7]. As highlighted in these reports, the absence of comprehensive, long-term planning for airports is poised to yield significant challenges in the near future. The primary culprit behind these impending issues is the inherent capacity limitations of airport runways [8].

Rather than being challenging, building new infrastructure is believed to be costly. For instance, the construction of just one runway at Dublin Airport costs about 320 million euros [7]. Thus, the optimal use of the existing infrastructure through optimization is deemed to be the best solution for the unresolved problem of airport congestion [9]. One of the significant operations in air traffic control is the landing procedure [8], which causes congestion and is one of the major reasons for delayed flights [9], which can cost the airline industry 505 billion dollars annually [10]. In addition, as it is clear, not only can the delays lead to passenger dissatisfaction and complaints about the airlines [11], but the environment will also be negatively affected by repeated delays [12] since delays can increase fuel consumption and air pollution [13].

The Aircraft Landing Problem (ALP) is one of the major issues in the field of air traffic scheduling and control [14, 15]. The most significant challenge of the ALP is allocating the airport's runways to aircraft and determining their landing time and order [16]. Since it is one of the main constraints in the ALP, the allowed distance between the aircraft during flight and landing has to be observed to have a safe landing. Based on the aircraft's speed, a time interval should be considered to avoid accidents [17]. Another vital and necessary constraint of the ALP is the use of time windows, according to which the aircraft is only allowed to land at the specified time among the earliest and latest allowed landing times [18]. The available methods and approaches to solve the ALP are divided into two main categories: (i) exact methods and (ii) approximate methods. Exact algorithms trace the best possible answer, including branch and bound, branch and cut dynamic programming [19]. Exact techniques, such as mixed-integer programming, have been employed in the context of ALP [20–23].

Furthermore, for approximate methodologies, meta-heuristic optimization strategies like Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Immunoglobulin-Based Artificial Immune System (IAIS), Grey Wolf Optimizer (GWO), and flower pollination algorithm (FPA) have been utilized. It is worth noting that GA-based solutions are more commonly favored [24]. Since finding the manual global optimum is quite time-consuming [25], and the complexity of the problem will increase as the number of aircraft and runways increases [19], the ALP can be classified as an NP-hard problem [26, 27]. Although exact algorithms can solve three instances with up to 50 aircraft without much computational effort, they have difficulty providing good solutions when a large number of aircraft is considered [28].

PSO was used by [29] to minimize the total penalty cost due to the deviation of aircraft landing times from the respective Target Landing Time (TLT). In [30], the authors used IAIS to minimize the cost of the flights. Authors in [31] proposed the GWO algorithm to evaluate three objective functions: the airport runway efficiency, the costs, and the fuel consumption. In [32], FPA was proposed to solve ALP while considering the delay objective. These are just a few examples of the many intelligent algorithms that make up the Swarm Intelligence approach, which is an approximation of an algorithm. Most of these works lack multi-runway mode consideration. Thus, they fail to reflect the current realities of the aviation industry as well as delivering more accurate and relevant analysis.

This study provides an alternative to deal with these situations. Approximate algorithms can tackle small-scale NP-hard problems with superior near-optimal solutions in a fraction of the time [33–35]. The improved Imperialist Competitive Algorithm (ICA) is applied to solve this problem on a larger scale. In addition to two historical research goals of minimizing aircraft landing delays and cutting down expenses related to early or late landings, we introduce an innovative objective simultaneously. Our focus is on achieving a balance between runway usage while also curbing expenses stemming from both early and late arrivals. The overarching aim is to prevent congestion in one lane while guaranteeing that the other lane doesn't remain underutilized, to use it optimally and more realistically. To tackle this modeling complexity, we employ the improved version of the ICA as our approach. In this paper, we have focused on some contributions simultaneously.

1. A novel approach for addressing the ALP with multiple criteria, employing a delay and early landing cost optimization technique and runway balance strategy, which enhances runway utilization, is presented.
2. The problem is tackled using a multi-runway solution strategy, which will reflect the current realities of the aviation industry and provide a more accurate and relevant analysis.
3. To enhance the population's diversity, the turbulent operator is applied to the proposed ICA since their limited searching ability makes it difficult to jump out of the local optimal when facing large-scale scheduling.

The remainder of the paper is organized as follows. Section 2 describes the objectives and constraints of the proposed model. Section 3 outlines the approach

employed for algorithm comparison and details the algorithms that were put into practice. The tested scenarios and the simulation results are described in Sect. 4; also, Sect. 5 is devoted to the computational results. The paper ends with the conclusion and future works in Sect. 6.

2 Proposed model

The purpose of this section is to define the basic concepts related to the scheduling of aircraft landings. First, we will define the types of constraints in this problem, and then we will point out the goals used in such problems.

2.1 Problem statement

A group of aircraft is approaching the terminal part of the airport. The goal of the ALP is to determine the order and landing time for all aircraft so that all constraints are satisfied and we reach the set goals. The ALP entails mapping each aircraft to a landing time to satisfy operational constraints while optimizing certain criteria. When an airport has more than one runway, controllers must choose which runway will be used for landing. This choice is based on several variables, including the layout of the airport and the direction of the incoming aircraft. The purpose of this section is to define the basic concepts of ALP. First, we define the types of constraints in this problem, and then we point out the goals used in such problems.

2.2 Constraints

ALP tries to schedule the landing times of arriving aircraft while taking into account several operational constraints. There are three main constraints that we must observe as hard constraints of the problem. In addition to hard constraints, there are some soft constraints of an aircraft landing, and in this section, we explain each separately.

2.2.1 Landing time windows constraint

Each aircraft appears at a time, which is called appearance time. For the landing time of each aircraft, there is a period in which the aircraft is not allowed to land before or after that period, which is called a time window. In other words, the landing time of the aircraft must be between the Earliest Landing Time (ELT) and TLT. Also, the minimum and maximum speed of the aircraft is one of the factors in determining the ELT and Latest Landing Time (LLT) of the aircraft [36].

The time window for the landing of the airplane n is, defined by $[E_n, L_n]$, which $E_n \leq G_n \leq L_n$, where E_n is the ELT, L_n is the LLT for the aircraft and G_n is the target (preferred) landing time for the aircraft. The aircraft landing process includes four important and clear time parameters, which include (i) appearance time, (ii) ELT, (iii) TLT, and (iv) LLT. A penalty is applied for each unit of delay in comparison with the

target time. Also, an early arrival penalty will be applied for each time unit of the early arrival of the airplane.

2.2.2 Separation time constraint

To maintain the aerodynamic stability of the airplane and for safety reasons, the minimum separation distance between the aircraft should be maintained and observed to prevent collision or damage occurring when two consecutive aircraft are landing. However, in some problems, instead of the separation time, the separation distance is considered. This can be converted into the distance by knowing the speed of the aircraft.

As shown in Fig. 1, s_{nm} indicates the separation time constraint (≥ 0) between airplane n landing and airplane $E_n \leq G_n \leq L_n$ landing (herein, the airplane n lands before the airplane m , $n \in (1, \dots, NA)$; $m \in (1, \dots, NA)$; $n \neq m$).

2.2.3 Creating a trade-off between runway use (runway balance strategy)

ALP is an NP-hard problem, and when dealing with big-size examples, the calculation amount is astronomically vast. Without an appropriate plan and load imbalance, another unforeseen phenomenon may occur. Here, a scheduling instance will be chosen to highlight the significance of loading balance. It is known as another new objective because it also deals with creating a balance between bands in this article. Due to the creation of unbalanced traffic in the bands and the optimal use of the bands, we used a new objective to balance the bands. This objective can prevent the occurrence of traffic in one band and the other band remaining empty.

2.3 Objectives

To solve the ALP, we used important and widely used objectives in the aviation industry. In the following, we will describe the objectives.

2.3.1 Delay

Air transportation is a very complex issue that has many stakeholders. Airports are largely dependent on passengers for income, so they need a lot of passengers to increase income, but they also need to reduce airport crowding [37]. Increasing congestion in airports will cause complexity in air traffic management as a result of increasing the delay in the landing of aircraft. The increase in the number of flights will cause traffic on the runways and make them busy. Besides, the next aircraft must wait to land, which will cause delays in the passengers' flights, costs, and environmental pollution.



Fig. 1 Separation constraint

Increasing the delay in the landing of aircraft will increase fuel consumption and cause pollution to the environment [38]. There is no doubt that delay is one of the most important and widely used objectives for the ALP because ignoring it will reduce air traffic control performance. For each aircraft, there is a time limit called target time, which means the ideal time of the aircraft's landing, and a time called landing time, which means the time of the aircraft's landing according to the plan. If the aircraft lands after the planned time, it will cause a delay for the passengers. In other words, the delay can be defined in Eq. (1).

$$z_1 = x_n - G_n \quad (1)$$

where z_1 presenting delay, G_n is target time, x_n is landing time. Delay is the most important objective for ALP, and many researchers have considered delay as the main objective of their problem (for example, minimizing the total delay [39] and minimizing the average delay [19]). There is an early landing as opposed to a late landing. If the plane lands earlier than the scheduled time, it means that it has early land. Equation (2) indicates the number of early arrivals of the plane.

$$z_2 = G_n - x_n \quad (2)$$

2.3.2 Cost

The cost has been a popular and traditional ALP objective, similar to the prior purpose. The cost is related to the early or late landing time of the aircraft. If the aircraft lands early, it causes the runway to be occupied, and the next aircraft is not allowed to land, and it causes the higher aircraft to consume more fuel and wait for the runway to be empty. If the aircraft lands late, it will cause a delay for the flight passengers. Hence, landing early or landing late imposes a cost on the airport. For the early and late landing of the aircraft, a penalty amount is considered for each time unit, and the total cost can be calculated using the following Eq. (3).

$$z = \begin{cases} f_n \times (G_n - x_n) & x_n \leq G_n \\ b_n \times (x_n - G_n) & G_n < x_n \end{cases} \quad (3)$$

where f_n is the penalty for the early landing of an aircraft, b_n is the penalty for a late landing, G_n is the target time, x_n is the landing time, and z is the cost. Various researchers, including [4, 40–43], have aimed to minimize the cost of deviation from TLT, which includes the cost of the early and late landing of aircraft.

3 Objective optimization for ALP

In this article, we considered cost objectives to solve the problem of aircraft landing time, which include (i) reducing the cost and increasing the efficiency of the runway by creating a balance between the runways. In the following, we will examine the existing objectives and constraints for this problem.

3.1 The mathematical formulation of ALP

As mentioned above, the ALP is finding the best landing time for a set of incoming aircraft so that the landing order of the aircraft and the landing time of the aircraft are close to the available optimal time. In other words, the time interval between the actual landing time and the optimal landing time should be minimized. Also, the distance between the aircraft should be observed to prevent air accidents. In this section, we will formulate the problem. Table 1 also mentions the notations and the decision variables.

In this article, we have used an important objective that researchers have always investigated. Equation (4) shows our fitness function.

$$z = \min \sum (b_n \times Z_2 + f_n \times Z_1) \quad (4)$$

Aircraft landing limitations involve a range of factors that impact when and how aircraft can land at airports. Two specific instances of these limitations are time windows and separation time. Both time windows and separation times play crucial roles within air traffic management systems, working to avert clashes, uphold safety, and enhance the sequencing of aircraft arrivals at airports. These restrictions are crafted to harmonize the requirement for smooth operations with the utmost priority of aviation safety. Equation (5) represents the first and most important constraint of the problem, time windows, which state that the landing time of the aircraft must be between the earliest allowed landing time and the latest allowed landing time of the aircraft. Equation (6) is employed for the computation of the disparity between the planned landing time and the TLT. Equation (7) limitation shows that two aircraft cannot land on the same runway at the same time, and one land before the other. Equation (8) guarantees the consistency of scheduled runways. Equation (9) ensures

Table 1 Notation and variable

Notation/variables	ALP
E_n	Earliest landing time of each aircraft
L_n	The latest landing time of each aircraft
f_n	Penalty for arriving early
x_n	Landing time
b_n	Penalty for arriving lately
G_n	Target landing time
y_{nm}	aircraft n lands before aircraft m
δ_{nm}	Whether aircraft n and m land on the same runway
γ_{nd}	Whether aircraft n is allocated to runway d
S_{nm}	Separation time units between two ordered aircraft, n , and m , landing on the same runway
s_{nm}	The separation time between aircraft n and aircraft m for those landing on the different runways
NR	Number of runways

that each aircraft lands on just one of the runways. Constraints in Eqs. (10) and (11) also indicate that the parameters used take only binary values. The constraint of separation time is handled by Eq. (12). Equation (13) shows the range of the decision variables. Table 1 is presented to show the parameters used in the above formulas. Equations (14)–(17) show the decision variables for ALP. Equation (18) shows the tardiness made by aircraft n and also Eq. (19) shows the early arrival of aircraft n .

$$E_n \leq x_n \leq L_n, \quad \forall n \in \{1, 2, 3, \dots, NA\} \quad (5)$$

$$x_n - G_n = \alpha_n - \beta_n, \quad \forall n \in \{1, 2, 3, \dots, NA\} \quad (6)$$

$$y_{nm} + y_{mn} = 1, \quad n \neq m \in \{1, 2, 3, \dots, NA\} \quad (7)$$

$$\delta_{nm} \geq \gamma_{nd} + \gamma_{md} - 1, \quad n \neq m \in \{1, 2, 3, \dots, NA\}, \\ d \in \{1, 2, 3, \dots, NR\} \quad (8)$$

$$\sum_{d=1}^{NR} \gamma_{nd} = 1 \quad n \in \{1, 2, 3, \dots, NR\} \quad (9)$$

$$y_{nm} \delta_{nm} \in \{0, 1\} \quad n \neq m \in \{1, 2, 3, \dots, NA\} \quad (10)$$

$$\gamma_{nd} \in \{0, 1\}, \quad n \in \{1, 2, 3, \dots, NA\}, \\ d \in \{1, 2, 3, \dots, NR\} \quad (11)$$

$$x_m \geq x_n + s_{nm} \delta_{nm} + s_{nm}(1 - \delta_{nm}) - M \delta_{mn} \\ \text{Where } M \text{ can be replaced by } (L_n + \max(s_{nm}, s_{mn}) - E_n) \\ n \neq m \in \{1, 2, 3, \dots, NA\} \quad (12)$$

$$x_n, \alpha_n, \beta_n \geq 0, \quad \forall n \in \{1, 2, 3, \dots, NA\} \quad (13)$$

$$\delta_{nm} = \delta_{mn} \quad (14)$$

$$\gamma_{nd} = \begin{cases} 1, & \text{if plane } n(n = 1, \dots, p) \text{ lands on runway } d(d = 1, \dots, RC) \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

$$y_{nm} = \begin{cases} 1, & \text{if plane } n \text{ lands before plane } m \\ 0, & \text{otherwise} \end{cases} \quad n \neq m \in \{1, 2, 3, \dots, NA\} \quad (16)$$

$$\delta_{nm} = \begin{cases} 1, & \text{if plane } m \text{ and plane } m \text{ lands in same runway} \\ 0, & \text{otherwise} \end{cases} \quad n \neq m \in \{1, 2, 3, \dots, NA\} \quad (17)$$

$$\alpha_n = \max(0, x_n - G_n) \quad (18)$$

$$\beta_n = \max(0, G_n - x_n) \quad (19)$$

4 Improved ICA for ALP

In recent times, a multitude of optimization challenges have been explored using the ICA due to its notable benefits, such as its rapid performance and precision in locating optimal solutions [44, 45]. The ICA method introduced by Hosseinia and Khaled in 2014 drew its inspiration from socio-political behaviors, initially devised for tackling continuous optimization problems [46]. This improved ICA approach can also be effectively employed in solving permutation-based optimization challenges like ALP. The schematic representation of the proposed ICA technique for addressing multi-objective ALP is depicted in Fig. 2, and its operational procedure can be encapsulated in the subsequent steps. The diagram in Fig. 2 illustrates the process of the suggested ICA for ALP.

Step 1: This phase involves setting up various improved ICA parameters, encompassing the maximum iteration count ($i_{\max}$), the size solutions (n_{pop}), the size of the imperialist group (n_{imp}), and the possibility of revolution). These parameters collectively ensure the viability of the potential solutions.

Step 2: Initialize the Empire. It is common knowledge that the commencement phase of an algorithm is its initial step. Consequently, the precise procedure for initialization will be detailed in this segment. As pointed out in Sect. 2.2, the ALP is bound by several constraints. Given this context, it becomes evident that a random initialization would yield numerous solutions that run counter to these constraints. As a result, these potential solutions would lack the quality necessary to develop into satisfactory scheduling solutions. Therefore, we used the method used in [32] for initialization and applied it to our improve-ICA. After the initialization of the imperialist and colonial entities has been completed, Eq. (20) has been used to calculate the total cost of the empire, which includes both the imperialist nation and its affiliated colonies:

$$Cemp_m = CImp_m + \mu \times \{mean(Ccol)\} \quad (20)$$

where $0 < \mu < 1$

where $CImp_m$ is the total cost of the m th empire, $Cemp$ is the cost of imperialism in the m th empire, and $mean(Ccol)$ is the mean of the cost of the colonies belonging to the m th empire. The constant parameter μ is the colonies' consideration rate, and it is usually set to be smaller than unity $\mu < 1$.

Step 3: Assimilate the Empire. In the classic ICA, assimilation and revolution are designed for Continuous data. So, we use uniform crossover for assimilation. We have used the uniform crossover for assimilation. For every colony, we have chosen random numbers of indexes randomly and have set the values of these indexes to the corresponding values in the imperialist.

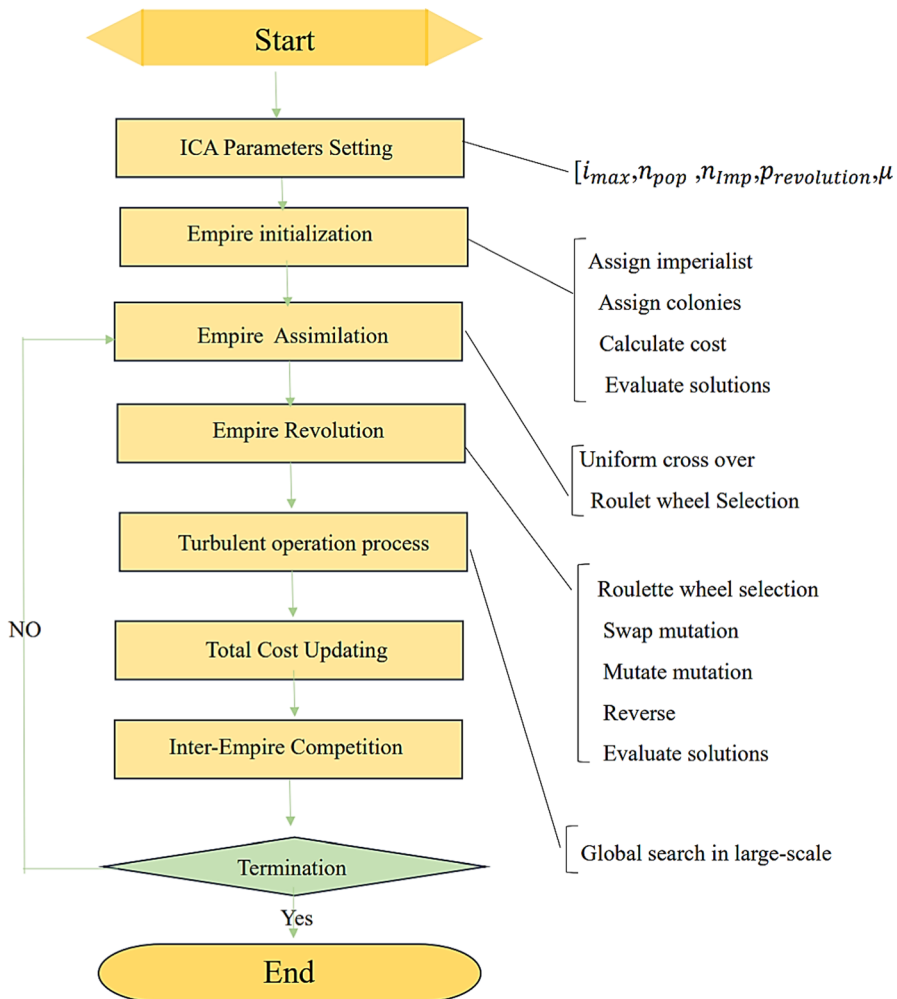


Fig. 2 Flowchart of improved ICA

$$X_i^{t+1} = \begin{cases} imp_j & rand_1 < rand_2 \\ X_{ij}^t & \text{Otherwise} \end{cases} \quad (21)$$

In Eq. (21) imp is the imperialist and X_{ij}^t is the index number j in the colony number i in the generation t .

Step 4: Revolutionize the Empire. For the revolution process, first, we generate a random number r . If $r < \frac{1}{2}$ then we change the value of a random index to a random value other than its current value. if $\frac{1}{2} < r < \frac{3}{4}$ then we choose random indexes and swap their values. Otherwise we choose two random indexes and reverse the list between them(including themselves). Equation (22) shows the revolutionized process.

$$X_i^{t+1} = \begin{cases} \text{mutate}(X_i^t, r_i) & r < \frac{1}{2} \\ \text{swap}(X_i^t, r_i, r_k) & \frac{1}{2} < r < \frac{3}{4} \\ \text{revers}(X_i^t, r_i, r_k) & \text{otherwise} \end{cases} \quad (22)$$

X_i^t is the colony number i in the generation number t . r_i and r_k are two random indexes.

Step 5: The competition within the Empire. This phenomenon happens when a colony gets a better solution in comparison with its imperialists. Here, the colony and the imperialist will be replaced.

Step 6: As the colony and its imperialists might be changed during the process, the total expenses should get an update based on Eq. 20.

Step 7: Within this step, which is known to be the most important in the ICA process, some weaker colonies will give their place to stronger colonies. However, if there was not a weak empire to be replaced by a strong colony, then the empire would collapse and change into a colony.

Step 8: This step is known as Termination, where there are two conditions. 1- the maximum number of iterations, and 2- there is only an empire left.

4.1 Improved ICA algorithm embedded with runway balance for ALP

The classic ICA is one of the famous and well-known algorithms for continuous optimization problems. Thus, to solve the ALP, which is a discrete problem, it is necessary to change this algorithm. Therefore, in the following, an improved method for ICA is proposed to solve the ALP problem.

4.2 Solution representation

It is necessary to create an appropriate encoding law for the potential solution. Here, a straightforward example is chosen to clarify the solution representation. Assuming that the selected instance has three runways that can be assigned and ten planes that need to be scheduled, the candidate solution should be a ten-dimensional vector with one, two, or three decision variables for each. A possible solution's representation can be shown in Fig. 3 if it is 1, 3, 3, 2, 2, 1, 2, 2, 1, 1.

This figure shows that Aircraft 1, 6, 9, and 10 are on Runway 1, aircraft 4, 5, 7, and 8 are on Runway 2, and Aircraft 2 and 3 are on Runway 3.

4.3 Runways initializations

In the classic ICA, the population is initialized randomly, which, for landing problems, generates many solutions that conflict with the constraints. These candidates lack the requisite quality to advance into schedules that align with the desired standards. During the initialization phase, multiple potential solutions are generated, taking into account runway factors, which p_s signifies the population size. For each solution, an initial value of zero should be assigned to the latest active time for every

1	3	3	2	2	1	2	2	1	1
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Fig. 3 Solution representation

runway. Additionally, the TLT of all aircraft needs to be arranged in ascending order. This arrangement ensures that the TLT of any given aircraft in the sorted sequence is later than that of the preceding one. The first aircraft n_1 lands on a runway randomly. For each two sorted aircraft n_k, n_{k+1} , if $G_{n_{k+1}} > G_{n_k} + S_{n_k, n_{k+1}}$, the aircraft n_{k+1} is assigned to the runway that n_k lands on, the last active time of the runway which n_{k+1} lands on should be updated, and its value should be set to $G_{n_{k+1}}$. Otherwise, the aircraft $G_{n_{k+1}}$ should be assigned to another runway with the shortest last active time, and the last active time of this runway should be set to $G_{n_{k+1}}$. Namely, aircraft n_k and aircraft n_{k+1} are assigned to different runways. This special process of initialization repeats until the P_s candidate solutions are generated.

4.4 Turbulent operation process (intra-empire competition)

When facing large-scale scheduling, their limited searching ability makes it difficult to jump out of the local optimal [32]. In other words, improved ICA, when facing large-scale scheduling, the limited searching ability makes it hard to jump out of the local optimum, so a turbulent operator is applied for every colony after assimilation and revolution. Thus, the turbulent operator is applied to the proposed ICA to enhance the diversity of the population. This turbulent process contains a swap operator and an insert operator [47]. Its particular process can be depicted in Figs. 4 and 5.

After the process is done, the result is shown in Fig. 5.

4.5 Runway balance strategy

As it is commonly understood, the task of scheduling aircraft landings presents a challenging problem known to be NP-hard. Particularly when the problem has a significant scale, the computational workload becomes exceptionally substantial [48]. Furthermore, an unforeseen issue of load imbalance could arise in the absence of a suitable strategy. To underscore the significance of achieving balanced loading, we will opt to choose a scheduling example for demonstration. Imagine a scenario

1	1	2	1	2	2	1	2	2	1
---	---	---	---	---	---	---	---	---	---

Fig. 4 Specific procedure of swap operator (before)

1	2	2	2	1	1	2	1	2	1
---	---	---	---	---	---	---	---	---	---

Fig. 5 Specific procedure of swap operator (after)

where 2,000 aircraft are getting ready to land at an airport that only possesses three runways. With a large number of aircraft but only a limited number of runways available, the allocation of aircraft to each runway could lead to a significant disparity in workload. For instance, if 1400 aircraft were to land on the first runway, 400 on the second, and 200 on the third, issues would arise. This is because, without considering an equitable distribution of aircraft across the runways, the current distribution isn't fair or balanced, resulting in congestion primarily on the first runway. Hence, a runway equilibrium approach is formulated herein to ensure an even distribution of the workload across each runway. Initially, the count of aircraft on each runway needs to be computed. Subsequently, an aircraft that has the highest count of landing aircraft is selected randomly from the runway. Concurrently, identification of the runway with the fewest aircraft to accommodate landings is carried out, and the selected aircraft is allocated for landing there. If the number of aircraft on runways with both the minimum and maximum counts is equivalent, then a runway is arbitrarily chosen. This sequence of actions persists until the variance in runway counts is minimized to less than one. The pseudocode of the runway balance strategy can be shown in Algorithm 1.

Algorithm 1: The pseudocode for making balance in runways

```

1.  function outcome = Runway_balance (answer)
2.      [maxrunway, maxCount] = searchforMaxRunway(answer);
3.      [minrunway, minCount] = searchforRunway(answer);
4.      while maxCount-minCount <= 1
5.          Randomly choose an airplane from the max runway and allocate it to the main runway.
6.          Update the answer;
7.          [maxrunway,maxCount] = findMaxRunway(answer);
8.          [minrunway, minCount] = findMinRunway(answer);
9.      End
10.  outcome = answer
11.  End

```

4.6 ICA pseudocode

The provided pseudocode in Algorithm 1 elucidates the sequential actions of the improved ICA algorithm. Through step 2, the initial segment, we address the process of initialization. Pertains to the Revolution process, and 4.4 encompasses the turbulent operation process. This loop continues until the specified condition is met, upon which the algorithm iterates toward the concluding stages once again. In other words, these steps are repeated until we reach max iteration (in this case, it is

considered equal to 5000(. The specific implementation steps of the provided ICA for ALP can be summarized in the pseudocode shown in Algorithm 2.

Algorithm 2: Improved ICA for Aircraft Landing Scheduling Problem

```

1.  X = Initpopulation (Ps);
2.  Emp = Initempires (X, Es);
3.  While T < Maxgen Do
4.    For I = 1:Es Do
5.      (Imperialist Revolution);
6.      Imp-Temp = Revolution (Emp[I].Imp);
7.      If Fitness (Imp-Temp) <= Fitness (Emp[I].Imp) Then
8.        Emp[I].Imp = Imp-Temp;
9.      End
10.   (Assimilate Colonies);
11.   Colonies-Temp = Assimilate (Emp[I].Colonies);
12.   (Revolution For Colonies);
13.   Colonies-Temp = Revolution (Colonies-Temp);
14.   (Turbulent Operator For Colonies);
15.   Colonies-Temp-Turb = Turbulent (Colonies-Temp);
16.   (Update Colonies Only If Cost Is Less);
17.   For C = 1:Len (Colonies-Temp) Do
18.     If Fitness (Colonies-Temp-Turb [C]) <= Colonies-Temp [C] Then
19.       Colonies-Temp [C] = Colonies-Temp-Turb [C]
20.     End
21.     If Fitness (Colonies, Emp[C]) <= Emp [I].Colonies[C] Then
22.       Emp [I].Colonies[C] = Colonies-Temp[C]
23.     End
24.   End
25. End
26. End

```

5 Simulation experiments and result analysis

5.1 Instance and experimental setup

Programming has been done with 15 small-scale scheduling cases and 17 large-scale scheduling cases chosen from OR-Library to validate the performance of the improved ICA. These instances of benchmarks are frequently used in the literature

[49, 50]. Throughout the experiments, Python was utilized to code all optimization algorithms and simulations; while, the experimental setup was executed on a machine equipped with an Intel Core i7-6700HQ quad-core processor clocked at 2.60 GHz, 16 GB of RAM and running the Windows 10 operating system.

5.2 Simulation experiments for small-scale scheduling

Fifteen small-scale scheduling examples from OR-Library are chosen to evaluate the performance of the ICA. These cases have undergone sufficient testing, where ICA performed 25 runs with a unique initial answer in each instance. In this part, since the optimal value for the dataset is available, the algorithm was executed until the optimal solution was reached, and then it ended. Also, the condition of max iteration is considered 5,000 as the end iteration, and if the desired algorithm fails to find the optimal solution up until the max iteration, then it will stop the program. The worst objective values, the best objective values, the mean objective values, and the standard deviations for these benchmarks are shown in Table 2. It also demonstrates the number of aircraft and runways and the optimum objective value for individual benchmark instances. According to the table, the great stability level of the proposed ICA for the aircraft landing scheduling issue is because the best, worst, and mean values of 25 separate runs for each individual scenario are all exactly the same, and for each benchmark, the SD equals zero.

Table 2 Statistical data of ICA for 15 small-scale scheduling cases

Instance	NA	NR	Opt	Best value	Worth value	Mean value	SD
Airland1	10	2	90	90	90	90	0
		3	0	0	0	0	0
Airland2	15	2	210	210	210.01	210	0
		3	0	0	0	0	0
Airland3	20	2	60	60	60	60	0
		3	0	0	0	0	0
Airland4	20	2	640	640	640	640	0
		3	130	130	130	130	0
Airland5	20	2	<u>650</u>	<u>900</u>	900	900	0
		3	170	170	170	170	0
Airland6	30	2	554	540	540	540	0
		3	0	0	0	0	0
Airland7	44	2	0	0	0	0	0
Airland8	50	2	<u>135</u>	130	130	130	0
		3	0	0	0	0	0

The values that were able to perform the same as the existing optimal value are shown with bold and the results that the algorithm was able to perform better than the existing optimal value are shown with an underline

Table 3 Top scheduling solution and the assignment of runways gained by improved ICA for Airland 1–2

Instance	NA	NR	Best solution	Assignment of runways
Airland1	10	3	3 2 3 3 2 3 2 1 1 1	RW1:8,9,10 RW2:2,5,7 RW3:1,3,4,6
Airland1	10	2	2 2 1 1 2 1 2 1 1 2	RW1:3,4,6,8,9 RW2:1,2,5,7,10
Airland2	15	3	3 3 1 2 1 3 1 2 3 2 1 2 1 2 3	RW1:3,5,7,11,13 RW2:4,8,10,12,14 RW3:1,2,6,9,15
Airland2	15	2	1 1 1 2 2 2 1 1 2 2 1 1 2 2 2	RW1:1,2,3,7,8,11,12 RW2:4,5,6,9,10,13,14,15

Table 4 Top scheduling result and the assignment of runways found by improved ICA for Airland 3, 4, 5

Instance	NA	NR	Best solution	Assignment of runways
Airland3	20	3	3 2 3 1 2 3 2 1 3 1 1 2 3 2 1 3 2 1 2 1	RW1:4,8,10,11,15,18,20 RW2:2,5,7,12,14,17,19 RW3:1,3,6,9,13,16
Airland3	20	2	1 2 2 2 2 2 2 1 1 2 2 1 1 1 2 2 1 1 1 1	RW1:1,8,9,12,13,14,17,18,19,20 RW2:2,3,4,5,6,7,10,11,15,16
Airland4	20	3	2 3 1 2 1 3 1 2 1 3 1 1 2 2 3 3 2 2 3 1	RW1:3,5,7,9,11,12,20 RW2:1,4,8,13,14,17,18 RW3:2,6,10,15,16,19
Airland4	20	2	2 2 1 1 1 1 1 2 2 2 1 1 2 2 1 2 2 1 2 1	RW1:3,4,5,6,7,11,12,15,18,20 RW2:1,2,8,9,10,13,14,16,17,19
Airland5	20	3	3 3 2 3 2 2 1 1 3 2 2 3 3 1 1 3 1 2 2 1	RW1:7,8,14,15,17,20 RW2:3,5,6,10,11,18,19 RW3:1,2,4,9,12,13,16
Airland5	20	2	2 2 1 1 2 1 2 2 2 1 2 1 2 1 1 1 2 1 2 1	

Table 3 represents two examples of small-scale data. In the first case, we have ten aircraft and three runways, so the sequence is 3, 2, 3, 3, 2, 3, 2, 1, 1, and 1, which means that aircraft number one will land on runway three and aircraft number 2 on runway two, and so on. The assignment of the scheduling issues is displayed in the final column of Table 3, 4, and 5.

In our extensive analysis, various renowned algorithms were employed, aiming to facilitate meaningful comparisons. To evaluate the performance of our enhanced ICA algorithm on a smaller scale, we benchmarked it against four prominent meta-heuristic algorithms: GWO, PSO, IIAIS, and FPA. Two distinct criteria were utilized to assess the efficacy of our model compared to established models. The first criterion focuses on execution time; while, the second criterion delves into evaluating

Table 5 Top scheduling result and the assignment of runways found by improved ICA for Airland 6, 7, 8

Instance	NA	NR	Best solution	Assignment of runways
Airland6	30	3	1 3 2 1 1 2 3 1 3 2 3 1 1 3 3 2 2 3 1 2 1 2 2 2 1 1 3 3 3 2	RW1: 1, 4, 5, 8, 12, 13, 15, 16, 19, 21, 25, 26
				RW2: 3, 6, 10, 16, 17, 20, 22, 23, 24, 30
				RW3: 2, 7, 9, 11, 14, 15, 28, 27
Airland6	30	2	1 2 2 1 2 1 2 2 1 1 2 1 1 2 2 2 2 1 2 1 1 2 2 1 1 2 1 1 2	RW1: 1, 4, 6, 9, 10, 11, 13, 14, 19, 21, 22, 26, 25, 28, 29
				RW2: 2, 3, 5, 7, 8, 12, 15, 16, 17, 18, 20, 23, 24, 27, 30
Airland7	44	3	3 1 3 2 2 3 2 3 1 2 2 3 1 2 3 3 2 1 2 1 2 3 2 1 3 3 3 3 2 3 1 2 3 1 1 1 2 1	RW1: 2, 8, 13, 18, 20, 22, 23, 25, 29, 37, 40, 41, 42, 44
				RW2: 4, 5, 7, 10, 11, 14, 17, 19, 21, 24, 26, 28, 25, 38, 43
Airland7	44	2	1 2 1 1 2 2 1 1 2 1 1 2 2 1 1 2 2 1 1 2 1 2 1 1 2 1 1 2 2 2 1 2 2 1 2 2 2	Rw3: 1, 3, 6, 8, 12, 15, 16, 29, 30, 31, 32, 33, 34, 36, 39
				RW1: 1, 3, 4, 7, 8, 10, 11, 13, 14, 17, 18, 20, 22, 24, 26, 27, 29, 31, 32, 35, 38, 41
Airland8	50	3	1 3 3 1 1 1 3 3 3 2 1 2 3 1 1 2 1 3 2 3 1 3 2 1 3 1 1 3 3 2 2 2 3 2 2 2 1 2 1 1 2 1 3 1 3 2 3	RW2: 2, 5, 6, 9, 12, 15, 16, 19, 21, 23, 25, 28, 30, 33, 34, 36, 37, 39, 40, 41, 42, 43, 44
				RW1: 1, 4, 5, 6, 11, 14, 15, 17, 21, 24, 26, 27, 40, 42, 43, 45, 47
Airland8	50	2	2 1 1 1 1 1 2 2 2 1 2 2 2 2 1 1 2 2 2 2 2 1 1 2 2 2 1 1 2 2 2 1 1 2 1 2 1 1 2 1 1 2 2 1 2 1	RW2: 10, 12, 16, 19, 23, 30, 31, 32, 33, 35, 37, 38, 39, 42, 44, 49
				RW3: 2, 3, 7, 8, 9, 13, 18, 20, 22, 25, 28, 29, 34, 36, 46, 48, 50
				RW1: 2, 6, 10, 16, 17, 18, 24, 25, 28, 29, 30, 31, 32, 36, 37, 39, 41, 42, 44, 45, 48, 50
Airland8	50	2	2 1 1 1 1 1 2 2 2 1 2 2 2 2 1 1 2 2 2 2 2 1 1 2 2 2 1 1 2 2 2 1 1 2 1 2 1 1 2 1 1 2 2 1 2 1	RW2: 1, 7, 8, 9, 11, 12, 13, 14, 15, 19, 20, 21, 22, 23, 26, 27, 33, 34, 35, 38, 40, 43, 46, 47, 49

“ u ”, which represents the disparity between the achieved algorithm value and the optimal value. As can be seen in Table 6, the new notations are:

- O: The value of the optimal objective
- X: Algorithm’s best value
- T: the average execution time in seconds
- U: The percentage of difference between the obtained value and the optimal value is obtained through the following Eq. (23).

$$U = 100 \times \left(\frac{X - op}{op} \right) \quad (23)$$

Table 6 represents the results of implementing the mentioned algorithms on a small-scale (10–50 aircraft). To validate and compare the performance of the algorithms, three criteria with the titles T, X, and U have been used. According to this table, it can be understood that the improved ICA method has performed much better than the other presented algorithms in 15 cases and has been able to obtain the optimal value. However, for 1 case (Airland 5 for 2-band scheduling),

Table 6 Evaluations of computational outcomes for small-scale aircraft landing scheduling case

Instance	NR	O	Improved ICA			GWO			PSO			IIAIS			FPA		
			X	T	U	X	T	U	N	T	U	X	T	U	X	Ta	U
Airland1	2	90	90	12	0	91	14	1	90	12	0	90	17	0	90	18	0
Airland1	3	0	0	11	0	0	15	0	0	11	0	0	14	0	0	21	0
Airland2	2	210	210	12	0	230	17	18	210	13	0	210	18	0	222	26	5
Airland2	3	0	0	13	0	10	14	ND	0	14	0	0	17	0	0	27	0
Airland3	2	60	60	14	0	77	17	29	240	15	300	60	16	0	60	19	0
Airland3	3	0	0	15	0	16	17	ND	0	16	0	0	17	0	0	22	0
Airland4	2	640	640	16	0	662	15	3	641	17	5	640	21	0	640	25	0
Airland4	3	130	130	17	0	236	22	2.8	220	18	69	183	22	40	130	21	0
Airland5	2	650	<u>900</u>	16	38	117	21	81	910	22	40	915	18	40	980	25	50
Airland5	3	170	170	12	0	587	20	245	230	21	35	220	19	29	181	26	64
Airland6	2	554	<u>540</u>	11	<u>-2.5</u>	120	21	116	900	19	62	554	26	0	700	31	26
Airland6	3	0	0	22	0	87	22	ND	0	25	0	0	23	0	0	26	0
Airland7	2	0	0	11	0	0	22	0	0	26	0	0	27	0	0	27	0
Airland7	3	0	0	14	0	0	22	0	0	21	0	0	21	0	0	27	0
Airland8	2	135	<u>130</u>	11	<u>-3.8</u>	540	23	300	550	29	20	184	27	36	194	24	41
Airland8	3	0	0	15	0	0	24	0	0	28	0	0	31	0	0	26	0

The values that were able to perform the same as the existing optimal value are shown with bold and the results that the algorithm was able to perform better than the existing optimal value are shown with an underline

there has been some difference, which is displayed with the U variable. Even in cases such as Airland 6 and Airland 8, for which the U was 2.5 and 3.8%, respectively, ICA has been able to produce a better value than the optimal value. The average execution time (T) is a crucial factor to consider when assessing an algorithm's effectiveness. By examining the statistical information in Table 6, it can be understood that the mean execution time of ICA is considerably fewer than that of FPA, IIAIS, PSO, and GWO, which shows the effectiveness of the ICA as a solution to the ALP schedules. The presented method has been able to reduce the cost of late arrival and early arrival by 3.8 and 2.5%, respectively, in Airland 6 and Airland 8 in two runway modes.

5.3 Simulation experiments for large-scale scheduling

In this section, the ICA algorithm has been employed to solve the problem on a large scale, where since the landing time is unknown, only the value of the max iteration has been used as the end condition. For each of the cases here, improved ICA was run 25 times separately, and the worst target values, best target values, standard deviations, and mean target values obtained by ICA are reported in Table 7. Based on the outcome reports, the improved ICA method has been able to perform better than other existing methods presented in this section. In Table 7 and 8, Vopt is the

Table 7 Statistical archives of ICA for large-scale cases

Instance	NA	NR	Vopt [FPA]	Best value	Worth value	Mean	SD
Airland9	100	2	513.26	482	501	491.5	9.5
		3	76.98	75.5	77	76.25	0.75
		4	0	0	0	0	0
Airland10	150	2	1425.6	1440.38	2180	1475.06	35.31
		3	238.12	210.3	215.5	212.9	2.6
		4	34.22	26.4	27.6	27	0.6
Airland11	200	2	1724.2	1580.21	1593.1	1586.655	6.445
		3	285.92	285.42	291.31	288.365	2.945
		4	54.53	54.53	55.1	54.815	0.285
		5	0	0	0	0	0
Airland12	250	2	2461	2249.5	2289.5	2269.5	24
		3	259.99	249.3	250.3	249.8	1.5
		4	2.44	2.2	3.13	2.66	1.465
Airland13	500	2	5452.5	4800.12	5010.32	49,055.22	18.1
		3	785.5	503.5	551.1	527.3	33.8
		4	107.6	80.1	85.1	82.6	5.5
		5	0	0	0	0	0

The values that were able to perform the same as the existing optimal value are shown with bold and the results that the algorithm was able to perform better than the existing optimal value are shown with an underline

Table 8 Evaluation of computational outcomes for large-scale aircraft landing scheduling case

Instance	NR	Vopt (FPA)	Improved-ICA			GWO			IIAIS			ABC		
			X	T	U	X	T	U	X	T	U	X	T	U
Airland9	2	513.26	482	48	-6.9	510.1	211	-0.6	545	61	6.2	500	91	-2.5
	3	76.98	75.62	44	-1.9	129.1	176	67.7	74.3	85	-3.4	81	85	8.2
	4	0	0	41	0	43	231	ND	0	54	0	9	71	ND
Airland10	2	1425.6	1440.3	51	1.03	1700.1	330	19.27	2180	97	47.3	1611	101	13.0
	3	238.12	210.3	56	-11	341.2	241	43.2	328	54	37.8	291	105	38.3
	4	34.22	26.4	50	-22	150.04	251	337.1	35.3	94	6.7	41	99	20.5
Airland11	2	1724.2	1580.60	61	-8.3	1689	196	-2.3	1982	140	14.7	1890	102	9.6
	3	285.92	285.42	57	-0.9	410.2	212	43.4	345	65	21.0	319	94	11.5
	4	54.53	54.53	44	0	120	251	120	124	43	127.3	85	76	57.4
Airland12	0	0	0	41	0	41.3	275	ND	41	51	ND	0	55	0
	2	2461	2249.5	68	-8.5	2291	297	&&	2856	64	16.4	2918	131	18.5
	3	259.9	249.3	61	-4.7	820.2	205	216.7	356	63	23.5	391	126	50.0
Airland13	4	2.44	2.2	63	-9.0	139.2	301	5604	31	67	1179.3	4.1	120	70.8
	2	5452.5	4800.22	81	-11	7398.5	279	35.3	5600	91	2.7	6172	140	13.2
	3	785.5	527.3	84	-35	958.1	313.2	21.91	893	101	13.3	895	129	14.01
	4	107.6	82.6	86	-25	491.2	289	358	591	95	452.3	519	101	385.1
	5	0	0	90	0	450	242	ND	41	65	ND	0	81	0

objective value of the best solution for each benchmark instance in which we utilize the best solution [32]. According to Table 7, among 17 cases that have been investigated, in 16 cases, the proposed algorithm has performed better than the existing optimal value, which is designated as bold.

Statistical results comparison for large-scale ALP cases is offered in Table 8. Impressively, our novel algorithm showcased remarkable superiority over the FPA algorithm across multiple scenarios. The article emphasized the algorithm's robustness, particularly in intricate cases such as "airlnad13" with "runway=3," where a substantial 35% enhancement in the cost function was achieved. This attainment underscored the algorithm's exceptional performance and adaptability, even in challenging scenarios.

The comparison outcomes are documented in Table 8, showcasing results for benchmark scenarios encompassing 100–500 aircraft. Upon examination of Table 8, it becomes evident that the superior solutions achieved by the improved ICA stand out in numerous extensive benchmark scenarios. Upon scrutiny of the indicator (U), it is observed that the percentage disparities from the optimal solution, as obtained by the improved ICA, consistently exhibit negative values across all test cases (excluding Airland 10 with two runways). This implies that the proposed algorithm consistently produces superior solutions compared to IIAIS, GWO, FPA, and artificial bee colony (ABC). In the remaining 16 cases, the percentage disparities of the improved ICA are consistently smaller than those of the alternative scheduling methods.

5.4 Discussion

In Sects. 5.2 and 5.3, extensive simulation experiments have been conducted. In Sect. 5.3, sixteen cases of small-scale aircraft landing scheduling are examined. Although the presented algorithm has been able to perform well on small scales, it has shown better efficiency on larger scales. As it is clear from Table 8, the proposed method has been able to reduce the costs by 35% (it has performed better than

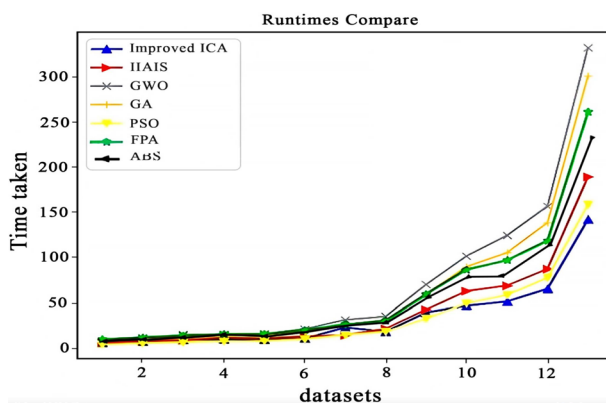


Fig. 6 Execution times in different datasets

the method presented in [32]). In other words, in all large-scale instances, the value of u is negative in all cases (except *airland10*), which means the reduction of costs compared to other algorithms. Various test outputs are offered extensively in seven tables. The mean execution times in Fig. 6 also indicate that the presented improved ICA converges faster. Considering that the ALP problem is complex and NP-hard, different approximate methods are used to solve it. In this article, to solve this problem, we used meta-heuristic algorithms, which are approximate. After comparing the presented method and the previous methods, we saw that the ICA algorithm has a very good ability to find the optimal solution in the shortest possible time. Therefore, improved ICA could be integrated into existing air traffic management systems. The results of 5 compared algorithms, including FPA, IIAIS, GWO, GA, PSO, ABC, and improved ICA, are shown in Fig. 7a–e. Due to the closeness of the results of the presented algorithms in small scales, they have been displayed only in higher dimensions. As can be derived from Fig. 7, when confronted with extensive scheduling tasks encompassing aircraft quantities spanning from 100 to 500, the suggested ICA algorithm distinctly exhibits its supremacy in computational efficiency. Furthermore, it substantiates that the enhanced version of ICA boasts an accelerated convergence rate compared to other methodologies. These findings underscore that in the face of more substantial challenges, ICA not only delivers noteworthy performance but also presents a capacity for swifter convergence.

Similar to other metaheuristics, ICA has both advantages and some unavoidable shortcomings. But some advantages encouraged us to use that. The comparison indicated that the ICA performs better than the other algorithms in terms of convergence rate because the ICA maintained a decreasing trend during its computation time. Also, according to [46], it has a minor chance of trapping into local optimal, and it is effective in the global search approach. A notable feature of ICA is the ease of performing neighborhood movements in both continuous and discrete search spaces; while, some other algorithms, such as PSO and ACO, have not been commonly used for continuous search space due to the difficulties of performing neighborhood movements.

As it is clear from Table 9, ICA is good in Convergence speed and computational time in comparison with other algorithms, which enlightens superior performance. While this approach is easy to implement, there are efforts for further research to develop a multi-objective ICA that can produce Pareto optimal solutions, utilizing techniques like ϵ constraint or Pareto sampling. Numerous real-world industrial domains grapple with large and intricate optimization problems that encompass diverse objectives. Table 10 provides the specific values of the parameters employed in their method.

5.5 Sensitivity analysis

We tested how the number of aircraft affects the computation times for problem instances with $|A| = 10, 20, 50, 200, 250$, and 500 aircraft, using a linear cost function. The other parameters remain unchanged. Table 11 shows the average and maximum computation times for 12 problem instances of each size. For $|A| = 200$ aircraft

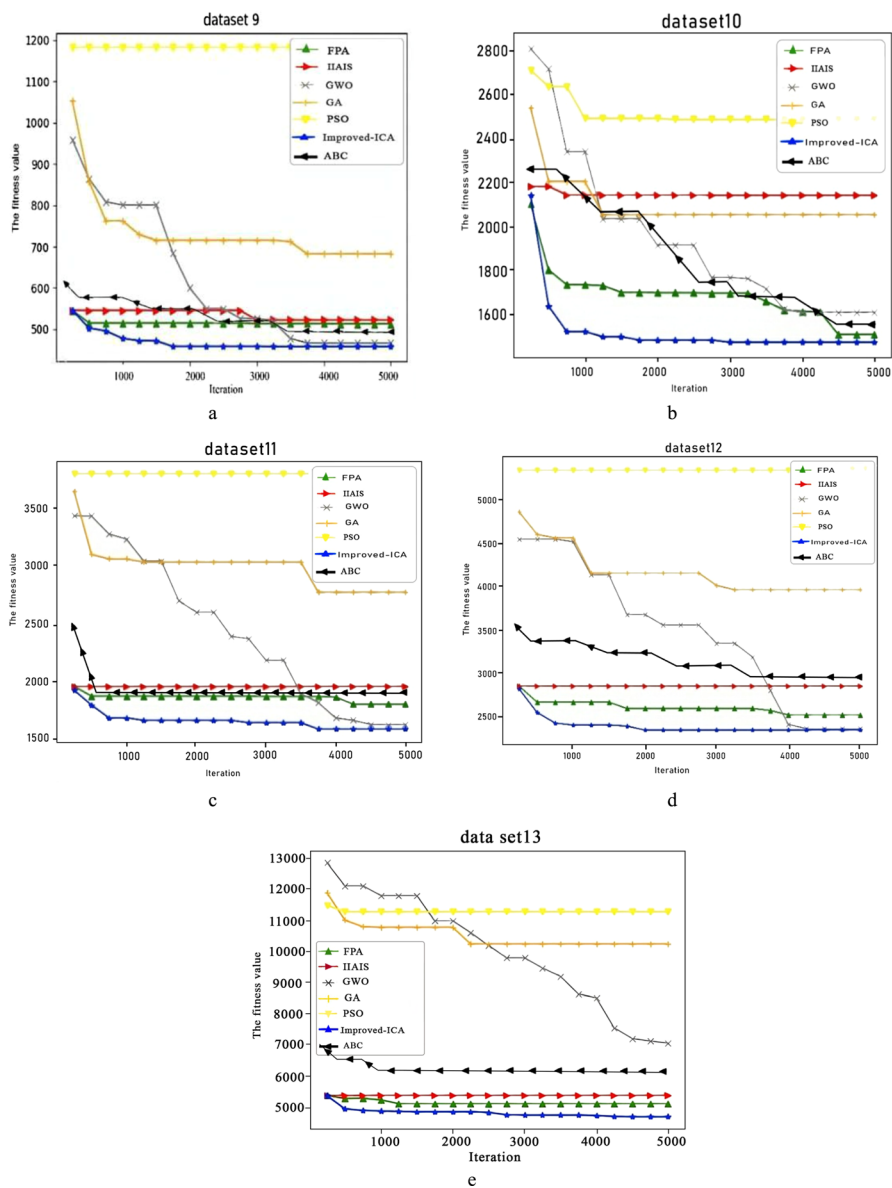


Fig. 7 Convergence curves of different Airlands with two runways (a-e)

Table 9 Comparison of some meta-heuristic algorithms

	GA	PSO	ICA	FPA
Convergence speed	slow	slow	medium	low
Computational time	high	medium	Low	medium
Local Search	weak	weak	Strong	strong

Table 10 Default values of key parameters

The key parameters	Value
Number of countries	800
Number of imperialist countries	80
Number of colonies	720
Coefficient associated	2
Revolution rate	0.3
Assimilation angle parameter	0.7
MAX-iteration	5000

Table 11 Sensitivity Analysis of times for the different problem sizes (in seconds)

Problem	NR=2		NR=3	
	Average	Maximum	Average	maximum
NA				
10	12	12	11	11
20	12	12	12	13
50	11	12	15	16
200	61	66	57	59
250	68	69	61	63
500	81	84	84	86

Table 12 Sensitivity Analysis of The Number of Runways in Cost

Problem	Airland 12		Airland 13	
	Average	Maximum	Average	maximum
NR				
2	2269.5	2289.5	49,055.22	5010.32
3	249.8	250.3	527.3	551.1
4	2.66	3.13	82.6	85.1

or less, the computation took less than 1 min in all of the test cases. The average computation time increases with the number of aircraft. But as it is clear from the results, only one minute and 20 s of execution time is needed to solve the problem with 500 aircraft.

Secondly, we tested how the number of runways affects the fitness value for problem instances with $|A|=250$ and 500 aircraft. As can be realized from Table 12, the increase in the number of runways has a direct relationship with the reduction of landing costs. And increasing the number of bands has a significant reduction in costs.

After implementing the enhanced ICA in the aforementioned model, we conducted a sensitivity analysis on the ICA parameters to assess their impact on further reducing costs in ALP. For this analysis, we selected three key ICA parameters: (1) the revolution rate, (2) the assimilation coefficient, and (3) the assimilation angle. In this sensitivity analysis, optimization runs were performed for each parameter while

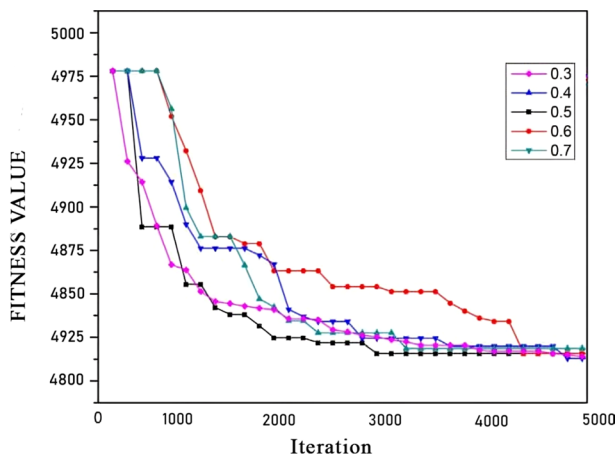


Fig. 8 Improved ICA performance with five different revolution rates for Airland 13

Table 13 Rankings of revolution rate parameters

Ranking	Value
1	0.3
2	0.4
3	0.5
4	0.6
5	0.7

keeping the others constant. Each parameter was divided into five sets of values, and 25 optimization runs were conducted for each value.

5.5.1 Revolution rate

The first ICA parameter subject to adjustment is the revolution ratio, which plays a pivotal role in inducing abrupt changes in the positions of a nation's colonies. A thorough examination of existing literature guided our choice of the recommended revolution ratio within the range of 0.3–0.7 [51]. Five distinct revolution ratio values (0.3, 0.4, 0.5, 0.6, and 0.7) were considered for evaluation. To mitigate the impact of stochastic variations in optimization, we conducted five optimization runs for each specified value. Subsequently, we compared and ranked the outcomes of the parameter sensitivity tests, relying on the total average objective value, as illustrated in Fig. 8 and Table 13. Our analysis identified the optimal revolution ratio as 0.3, suggesting that a moderate revolution ratio fosters a harmonious equilibrium between exploring and exploiting the solution space, thereby enhancing overall optimization performance.

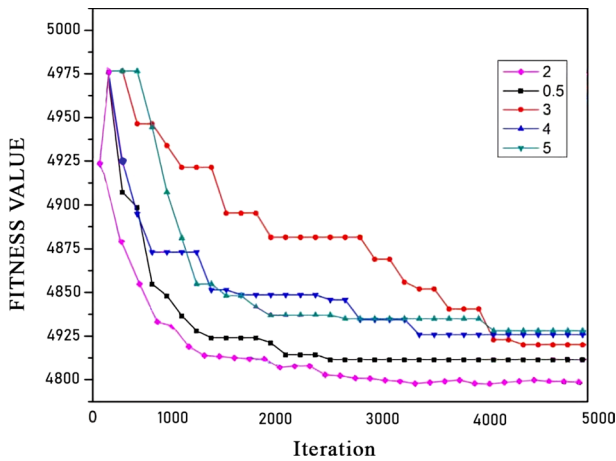


Fig. 9 ICA performance with five different assimilation coefficients for Airland 13

Table 14 Rankings of evolution rate parameters

Ranking	Value
1	2
2	0.5
3	3
4	4
5	5

5.5.2 Assimilation coefficient parameter

The next ICA parameter subjected to tuning was the assimilation coefficient, a key factor in the assimilation process governing the movement of colonies toward their imperialists. Employing a methodology akin to that applied for the revolution ratio parameter, we investigated five distinct assimilation coefficient values (0.5, 2, 3, 4, and 5) and conducted 25 optimization runs for each value. Subsequently, we compared and ranked the outcomes of the parameter sensitivity tests, relying on the total average objective values, as depicted in Fig. 9 and Table 14. The analysis revealed that the optimal assimilation coefficient value is 2.

5.5.3 Assimilation angle parameter

In this sensitivity analysis, the final ICA parameter under consideration was the assimilation angle, serving as a deviation in the assimilation process to ensure a wide search space and, consequently, enhance search diversification. We investigated five distinct assimilation angles (0.3, 0.5, 0.6, 0.7, and 0.8) and conducted 25 optimization runs for each value in the single vertical case. Subsequently, we compared and ranked the results

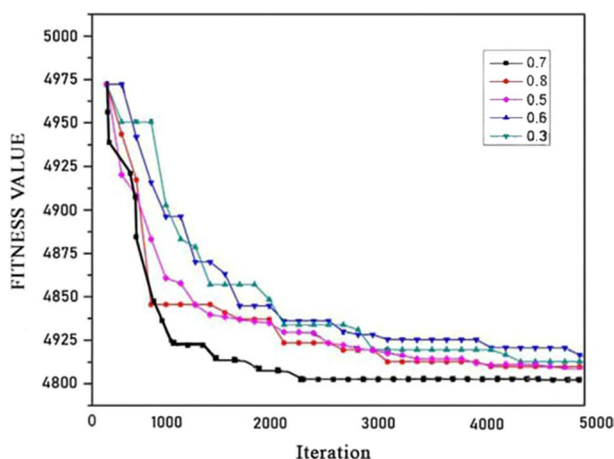


Fig. 10 ICA performance with five different assimilation angles for the Airland 13

Table 15 Rankings of revolution rate parameters

Ranking	Value
1	0.7
2	0.8
3	0.5
4	0.3
5	0.6

of the parameter sensitivity test based on the total average objective values, as illustrated in Fig. 10 and Table 15. The optimal assimilation angle value was identified to be 0.7.

6 Conclusions

This paper focuses on addressing the challenging issue of aircraft landing scheduling problems. The significance of this problem stems from its popularity and inherent difficulty. Since solving this problem is NP-hard, our proposed approach employed the Improved ICA to tackle the ALP. The primary objective of this study is to achieve equitable distribution of aircraft weight across runways. The study employs 32 benchmark instances of ALP, sourced from the OR-Library, to comprehensively evaluate the performance of the proposed scheduling algorithm. This evaluation involves comparing the outcomes obtained from ICA with those derived from other algorithms such as IIAIS, QWO, and more. The experimental results indicate that improved ICA outperforms alternative methods in both solution quality and computational speed. The complexity of the problem is confirmed as NP-hard. Introducing ICA, a novel population-based approach known

for its proficient exploration and exploitation capabilities, is considered. A modified version of ICA is introduced to address the ALP. The application of the turbulent operator further enriches population diversity and global search efficacy. A runway balancing technique is implemented to ensure balanced runway usage. Validation of the proposed ICA's effectiveness and efficiency involves testing it on 32 benchmark ALP instances from the OR-Library. The simulation results derived from improved ICA are benchmarked against outcomes from GWO, PSO, RBFPA, and other methods. The findings affirm that improved ICA surpasses these methods in terms of both solution quality and computational efficiency. The proposed method for solving the ALP using the ICA demonstrates promising results, but like any research, it has certain limitations that should be acknowledged. The proposed method for solving the ALP using the improved ICA demonstrates promising results, but like any research, it has certain limitations that should be acknowledged. There are some potential limitations and suggestions for addressing them in future work. The study may not fully capture the complexity of real-world airport operations, considering factors such as varying weather conditions, air traffic management policies, and dynamic changes in aircraft schedules. Future research could incorporate more realistic and dynamic factors into the model to make it more applicable to real-world scenarios. This might involve considering uncertainties in arrival times, dynamic changes in aircraft priorities, and other external factors.

Author contributions Kimia Shirini contributed to conceptualization, data curation, formal analysis, investigation, methodology, software, validation, visualization, and writing—original draft. Hadi S. Aghdasi contributed to conceptualization, data curation, formal analysis, investigation, methodology, project administration, supervision, software, validation, visualization, writing—review and editing. Saeed Saeedvand contributed to conceptualization, formal analysis, investigation, methodology, supervision, software, validation, writing—review and editing.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

References

1. Sabar NR, Kendall G (2015) An iterated local search with multiple perturbation operators and time varying perturbation strength for the aircraft landing problem. *Omega* 56:88–98
2. Diamantis Z, Tsahalis D, Borchers I (2002) Optimization of an active noise control system inside an aircraft, based on the simultaneous optimal positioning of microphones and speakers, with the use of a genetic algorithm. *Comput Optim Appl* 23:65–76
3. Su J et al (2023) A large neighborhood search algorithm with simulated annealing and time decomposition strategy for the aircraft runway scheduling problem. *Aerospace* 10(2):177
4. Câmara Á, et al (2016) A comparative study of meta-heuristics for the aircraft landing scheduling problem. In: 2016 11th Iberian Conference on Information Systems and Technologies (CISTI), pp. 1–4

5. Bertsimas D, Frankovich M (2016) Unified optimization of traffic flows through airports. *Transp Sci* 50:77–93
6. Sun X et al (2023) Air transportation and COVID-19: a tale with three episodes. *Transport Economics and Management* 1:13–21
7. Ng KKH et al (2017) Robust aircraft sequencing and scheduling problem with arrival/departure delay using the min-max regret approach. *Transp Res Part E Log Transp Rev* 106(1):115–136
8. Velilani K, Subramanya V, Chowdaiah P (2022) An algorithm for efficient real-time airplane landing scheduling. *ECS Trans* 107(1):69
9. Fan Y, Kalaba RE, Moore JE (2005) Arriving on time. *J Optim Theory Appl* 127:497–513
10. Bojanowski L, Harikiopoulou D, Neogi NA (2011) Multi-runway aircraft sequencing at congested airports. In: *Proceedings of the 2011 American Control Conference*, pp 2752–2758
11. Liang M, Delahaye D, Maréchal P (2017) Integrated sequencing and merging aircraft to parallel runways with automated conflict resolution and advanced avionics capabilities. *Transp Res Part C—Emerging Technol* 85:268–291
12. Shambour MK, Abu-Hashem MA (2023) Optimizing airport slot scheduling problem using optimization algorithms. *Soft Comput* 27(12):7939–7955
13. Visser HG (1991) Terminal area traffic management. *Prog Aerosp Sci* 28(4):323–368
14. Mokhtarimousavi S, Rahami H, Kaveh A (2015) Multi-objective mathematical modeling of aircraft landing problem on a runway in static mode, scheduling and sequence determination using NSGA-II. *Int J Optim Civil Eng* 5(1):21–36
15. Dönmez K, Çetek C, Kaya O (2022) Air traffic management in parallel-point merge systems under wind uncertainties. *J Air Transp Manag* 104:102268
16. Dönmez K, Çetek C, Kaya O (2022) Aircraft sequencing and scheduling in parallel-point merge systems for multiple parallel runways. *Transp Res Rec* 2676(3):108–124
17. Ng K, Lee C (2016) A modified variable neighborhood search for aircraft landing problem. In: *2016 IEEE International Conference on Management of Innovation and Technology (ICMIT)*. IEEE
18. Haouari M et al (2011) Exact approaches for integrated aircraft fleetling and routing at TunisAir. *Comput Optim Appl* 49:213–239
19. Balakrishnan H, Chandran BG (2010) Algorithms for scheduling runway operations under constrained position shifting. *Oper Res* 58(6):1650–1665
20. Beasley JE et al (2000) Scheduling aircraft landings—the static case. *Transp Sci* 34(2):180–197
21. Faye A (2015) Solving the aircraft landing problem with time discretization approach. *Eur J Oper Res* 242(3):1028–1038
22. Prakash R, Piplani R, Desai J (2018) An optimal data-splitting algorithm for aircraft scheduling on a single runway to maximize throughput. *Transp Res Part C: Emerging Technol* 95:570–581
23. Bartholomew-Biggs MC, Parkhurst SC, Wilson SP (2002) Using DIRECT to solve an aircraft routing problem. *Comput Optim Appl* 21:311–323
24. Zhang J, et al (2023) Dynamic optimal scheduling model of multi runway flight arrival and departure based on improved genetic algorithm. In: *2023 International Conference on Networking, Informatics and Computing (ICNETIC)*. IEEE.
25. Kjenstad D, et al. (2013) Integrated surface and departure management at airports by optimization. In: *2013 5th International Conference on Modeling, Simulation and Applied Optimization (ICMSAO)*, pp 1–5
26. Ernst AT, Krishnamoorthy M, Storer RH (1999) Heuristic and exact algorithms for scheduling aircraft landings. *Netw: Int J* 34(3):229–241
27. Anosri S, Panagant N, Champasak P et al (2023) A Comparative Study of State-of-the-art Metaheuristics for Solving Many-objective Optimization Problems of Fixed Wing Unmanned Aerial Vehicle Conceptual Design. *Arch Computat Methods Eng* 30(6):3657–3671
28. Gao J, Zhu X, Zhang R (2023) Optimization of parallel test task scheduling with constraint satisfaction. *J Supercomput* 79(7):7206–7227
29. Girish B (2016) An efficient hybrid particle swarm optimization algorithm in a rolling horizon framework for the aircraft landing problem. *Appl Soft Comput* 44:200–221
30. Vincent FY et al (2021) An improved immunoglobulin-based artificial immune system for the aircraft scheduling problem with alternate aircrafts. *IEEE Access* 9:16532–16545
31. Teimoori M et al (2021) A multi-objective grey wolf optimization algorithm for aircraft landing problem. *J Appl Res Ind Eng* 8(4):386–398

32. Zhou G, Wang R, Zhou Y (2018) Flower pollination algorithm with runway balance strategy for the aircraft landing scheduling problem. *Clust Comput* 21:1543–1560
33. Cheng Y, Cao Z, Zhang X et al (2023) Multi objective dynamic task scheduling optimization algorithm based on deep reinforcement learning. *J Supercomput*. <https://doi.org/10.1007/s11227-023-05714-1>
34. Tian Z et al (2023) Dynamic energy-efficient scheduling of multi-variety and small batch flexible job-shop: a case study for the aerospace industry. *Comput Ind Eng* 178:109111
35. Tian Z, Jiang X, Tian G, Li Z, Liu W (2023) Knowledge-based lot-splitting optimization method for flexible job shops considering energy consumption. *IEEE Trans Autom Sci Eng*. <https://doi.org/10.1109/TASE.2023.3303915>
36. Phirouzabadi AM, Aminnayeri M, Mahmoudian M (2013) Aircraft landing scheduling based on unavailability of runway constraint through a time segment heuristic method. *Int J Informat Commun Technol* 2:175–182
37. Zhang J et al (2020) Criteria selection and multi-objective optimization of aircraft landing problem. *J Air Transp Manag* 82:101734
38. Chen H, Solak S (2015) Lower cost arrivals for airlines: Optimal policies for managing runway operations under optimized profile descent. *Prod Oper Manag* 24(3):402–420
39. Rezaei H, et al. (2018) An iterative bidding approach applied to cost reduction in the context of aircraft landing problem. In: 2018 IEEE 22nd International Conference on Computer Supported Cooperative Work in Design (CSCWD). IEEE
40. Vadlamani S, Hosseini S (2014) A novel heuristic approach for solving aircraft landing problem with single runway. *J Air Transp Manag* 40:144–148
41. Ahmadian M.M, Salehipour A (2022) Heuristics for flights arrival scheduling at airports. *Int Trans Oper Res* 29(4):2316–2345
42. Lieder A, Stolletz R (2016) Scheduling aircraft take-offs and landings on interdependent and heterogeneous runways. *Transp Res Part E: Log Transp Rev* 88:167–188
43. Awasthi A, Kramer O, Lässig J (2013) Aircraft landing problem: an efficient algorithm for a given landing sequence. In: 2013 IEEE 16th International Conference on Computational Science and Engineering. IEEE
44. Pan Z, Lei D, Zhang Q (2018) A new imperialist competitive algorithm for multiobjective low carbon parallel machines scheduling. *Math Probl Eng* 2018:1–13
45. Sedighizadeh M, Eisapour-Moarref A (2017) The imperialist competitive algorithm for optimal multi-objective location and sizing of DSTATCOM in distribution systems considering loads uncertainty. *INAE Lett* 2:83–95
46. Hosseini S, Al Khaled A (2014) A survey on the imperialist competitive algorithm metaheuristic: implementation in engineering domain and directions for future research. *Appl Soft Comput* 24:1078–1094
47. Daley A et al (2012) Measuring entanglement growth in quench dynamics of bosons in an optical lattice. *Phys Rev Lett* 109(2):020505
48. Perboli G et al (2015) Flights and their economic impact on the airport catchment area: an application to the Italian tourist market. *J Optim Theory Appl* 164:1109–1133
49. Pinol H, Beasley JE (2006) Scatter search and bionomic algorithms for the aircraft landing problem. *Eur J Oper Res* 171(2):439–462
50. Salehipour A, Modarres M, Naeni LM (2013) An efficient hybrid meta-heuristic for aircraft landing problem. *Comput Oper Res* 40(1):207–213
51. Atashpaz-Gargari E, Lucas C (2007) Imperialist competitive algorithm: an algorithm for optimization inspired by imperialistic competition. In: 2007 IEEE Congress on Evolutionary Computation. IEEE

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applicable law.

Authors and Affiliations

Kimia Shirini¹ · Hadi S. Aghdasi¹ · Saeed Saeedvand²

✉ Hadi S. Aghdasi
aghdasi@tabrizu.ac.ir

Kimia Shirini
k.shirini@tabrizu.ac.ir

Saeed Saeedvand
saeedvand@ntnu.edu.tw

¹ Faculty of Electrical and Computer Engineering, University of Tabriz, Tabriz, Iran

² Department of Electrical Engineering, National Taiwan Normal University, Taipei, Taiwan